

- 6 (a) Structures and compounds involved in the light dependent stage of photosynthesis are listed, **A** to **J**.

A – thylakoid membrane

F – NADP

B – starch grain

G – electron

C – chlorophyll a

H – proton

D – chlorophyll b

I – ATP

E – water

J – chloroplast envelope

Complete Table 6.1 by matching each description with **one** letter chosen from **A** to **J** to show the correct structure or compound.

You may use each letter once, more than once or not at all.

Table 6.1

description	letter
accessory pigment	
location of ATP synthase	
acts as reaction centre	
transports hydrogen atoms	
diffuses through ATP synthase	
broken down in photolysis	

[6]

- (b) GP (PGA) and TP (triose phosphate) are intermediates of the Calvin cycle of the light independent stage. Some GP and TP are used for the synthesis of organic compounds.

- (i) Name a polysaccharide synthesised as a result of TP production.

..... [1]

- (ii) Name the additional element that is required for the production of amino acids from Calvin cycle intermediates.

..... [1]

[Total: 8]